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ORIGINAL ARTICLE

A novel splicing mutation in Marfan syndrome

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Abstract

Background Marfan syndrome (MFS) is a connective tissue disease involving multiple organs and systems such as cardiovascular, skeletal, and ocular systems and is also an autosomal dominant inheritance disorder.

Method A 30-year-old woman was rushed into the hospital owing to sudden persistent pain in the abdomen and died suddenly 2 days later. To find the real cause of death, a forensic autopsy was conducted owing to suspected medical malpractice, and the diagnosis of MFS was made in accordance with the 2010 revised Ghent nosology. By sequencing the gene of Marfan, aneurysm, and related disorders, a novel splicing mutation in the fibrillin-1 gene (FBN1) was detected. For the clinical characteristic findings (wrist and thumb sign) of the daughter, we recommend genetic analysis for the family. To better understand the role of the variant in the disease, we also investigated functional validation of this mutation.

Results According to the autopsy findings, the cause of death was acute cardiac tamponade caused by aortic rupture. DNA sequencing revealed a novel splicing mutation, c.5672-2delA, which was also detected in her daughter (II2). The functional validation of this mutation showed the base deletion at the same site in the PCR products using cDNA as a template. It is suggested that this mutation may cause abnormal spliceosome during transcription and may encode abnormal protein.

Conclusion A novel pathogenic splicing mutation (c.5672-2delA) was confirmed. Present work enriches the profile mutations in FBN1 associated with MFS and stresses the importance of postmortem genetic analysis in such cases.

Keywords Marfan syndrome · FBN1 · Splicing mutation · Autopsy · Sudden death

Introduction

Marfan syndrome (MFS) is a connective tissue disease involving multiple organs and systems such as cardiovascular, skeletal, and ocular systems and is also an autosomal dominant inheritance disorder [1]. The estimated prevalence rate of MFS was 1/5000 in Europe whereas 1–2/10,000 in China with no sex or ethnicity biases [2–4]. Although the clinical manifestations of MFS patients are variant, cardiovascular system lesions are still the leading cause of morbidity and mortality, especially thoracic aortic aneurysm and dissection (TAAD) [1]. It is reported that acute aortic dissection has an hourly mortality rate of 1 to 2% after symptoms in untreated patients [5]. Up to date, the life expectancy of MFS has increased to be close to normal. The reasons for the increase may include the benefits of cardiovascular surgery, as well as an increase due to the frequency of diagnosis of mild cases [6].

The gene encoding fibrillin-1, known as FBN1, is located on chromosome 15 [7]. According to previous studies, almost all the MFS can attribute to the mutation in FBN1 gene [8]. Up to date, more than 3000 mutations have been identified associated with MFS, among which 1847 different variants were

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collected (http://www.umd.be/fbn1/). Fibrillin-1 forms extracellular microfibrils, thus inevitably affect its structure and function. The variants in FBN1 could constitute abnormal microfibril, resulting in malformation of fibrillin and thus affecting the stability of connective tissues.

Here, a young female whose cause of death was acute aortic rupture finally diagnosed as MFS by forensic autopsy. Through genetic analysis, an unreported inherited splicing mutation (c.5672-2delA) in the fibrillin-1 gene (FBN1) was detected in the deceased and her daughter. The splicing mutation detected in intron46 could translate into abnormal mRNA splicing which encodes abnormal protein.

Material and methods

Clinical summary of patient

A 30-year-old woman was rushed into the hospital owing to sudden persistent pain in the abdomen. The pain was aggravated when lying down and relieved when sitting. The diagnosis of the pain was uncertain; hence, this patient received gastric acid suppression and symptomatic treatment to alleviate the symptoms. Two days later, the patient suddenly died whose echocardiography showed pericardial effusion. The final diagnosis was cardiac tamponade attributed to suspected aortic rupture.

Genetic analysis

Genomic DNA was prepared from muscle and used as the test material, the DNA was interrupted to prepare the library, then the DNA of the target gene coding region and the nearby shearing region are captured and enriched by the chip, and then the mutation is detected by the Sanger sequencing platform. Data interpretation rules refer to the relevant guidelines of American College of Medical Genetics and Genomics. Mutations were named according to the nomenclature recommended by the Human Genomic Variation Society.

Extension of the investigation to the family of the deceased

Considering the risk of the first-degree relatives of the deceased patient affected by MFS and at a risk of TAAD, we investigated the family throughly. We also collected the relative information: the deceased was adopted many years ago, so her biological parents' information was absent. The deceased had a son and a daughter, aged 10 and 3.5, respectively, we found that the daughter of the deceased showed wrist and thumb sign, and related information of the deceased and her daughter was collected in Table 1. We recommended genetic testing for her husband and children and acquired her husband's consents.

Table 1 Clinical information of the affected individuals in present family

Characteristic	Proband (I:2)	Proband's daughter (II:2)
Age	30	3.5
Gender	F	F
Wrist and thumb sign	+	+
Pectus excavatum	–	–
Plain flat foot	+	+
Reduced elbow extension	+	–
Facial features	+	–
Myopias	+	+
Retinal detachment	+	–
Height (cm)	176	106
Arm span (cm)	Unknown	106
AS/H	Unknown	1
Overgrowth of the long bones	+	–
Aortic dissection	+	Unknown

Functional validation of the mutation

Total RNA was isolated from the heart tissue from the deceased using UNIQ-10 Column Trizol Total RNA Isolation Kit (B511321, Sangon Biotech, Shanghai, China). To identify the quality of total RNA, RNA electrophoresis (1.5% agarose, 1 × TAE electrophoresis buffer) was performed. The first strand of cDNA was synthesized by M-MuLV Reverse Transcriptase (B500517, Sangon Biotech, Shanghai, China) using total RNA as template. Then polymerase chain reaction was conducted using cDNA with 2 pairs of primers designed according to the sequence near the mutation (Table 2) (LA Taq, DRR02AG, TaKaRa, Japan). Products were verified by agarose gel electrophoresis using SanPrep Column DNA Gel Extraction Kit (B518131, Sangon Biotech, Shanghai, China) to collect the products for Sanger sequence (this work was performed by Sangon Biotech, Shanghai, China).

Results

Autopsy findings

The deceased was 176 cm, with specific long legs and arms, and the joints of the fingers and toes enlarged as arachnodactyly. Special facial features such as disproportionately long and narrow head, recession of the eyeball within the orbit, and retrognathia were noted. Skin stria was located at the upper arm, axillary region, or thigh. Elbow extension reduced as the angle between the upper and lower arm measures 165° upon full extension. Approximately 650 g of blood and clot were found in the pericardial cavity. A 10.5-cm horizontal

Table 2 Primer sequences of PCR products of cDNA amplification

Amplicon	Forward primer 5′ → 3′	Reverse primer 5′ → 3′
FBN1-exon-47	AGGGATGACTGCTGCTGGAG	CACACGCACCTATACAGTCATTG

annular rupture of the internal layer was discovered 2.0 cm distal from the aortic valve (Fig. 1(a)). A 1.0-cm horizontal rupture of the internal layer of the right brachycephalic trunk was found 6.5 cm distal from the orifice of coronary sinus, 1.5 cm above which a 0.5 cm horizontal rupture of the internal layer of aorta was noticed 6.5 cm from the junction of ascending aorta and descending aorta (Fig. 1(b), (c)). A dissection was found in the aortic wall, and the blood had found its way via the outer membrane of the aorta which was 2.0 cm distal from the right atrial appendage and through the membrane into the pericardial.

In HE staining, cystic mucosal degeneration is visible in the middle of the aorta which stained blue in alcian blue stain.

According to the autopsy findings, the cause of death was acute cardiac tamponade caused by aortic rupture. Traumatic aortic rupture without atherosclerosis which occurs in young patients should consider hereditary diseases; considering the patient in our case, we conducted genetic testing to find the real cause behind the aortic disease.

Genetic analysis results

DNA sequencing revealed a novel splicing mutation, c.5672-2delA (Fig. 2). This novel splicing mutation, which was heterozygous, was located in intron46 of the FBN1 gene. This mutation was absent in multiple population databases, including the 1000 Genomes Project (https://www.ncbi.nlm.nih.gov/variation/tools/1000genomes/), dbSNP (http://www.ncbi.nlm.nih.gov/snp), the International HapMap Project (http://http.ncbi.nlm.nih.gov/hapmap/), the NHLBI GO Exome Sequencing Project (http://evs.gs.washington.edu/EVS/), and gnoMAD (http://gnomad.broadinstitute.org). In accordance with the suggestions of the American College of Medical Genetics (ACMG), because it is classified as PVS1, PM2, and PP4, the mutation is grouped as a "likely pathogenic variant."

Extension of the investigation to the family of the deceased

The heterozygous c.5672-2delA was also detected in her daughter (II2) and absent in her husband and son (I1 and II1).

Functional validation of the mutation

The PCR products using cDNA as a template were sequenced, and the base deletion at the same site was found (Fig. 3). The intron where the mutation is located is retained in the mRNA. It is suggested that this mutation may cause abnormal spliceosome during transcription and may encode abnormal protein.

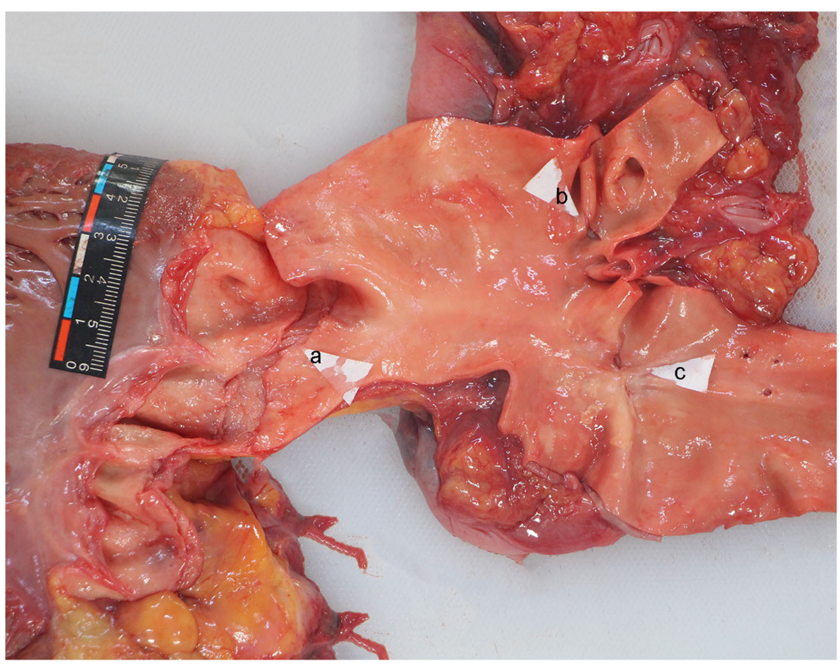


Fig. 1 Ruptures of the internal layer of aorta: (a) horizontal annular rupture in ascending aorta, (b) horizontal rupture in the right brachycephalic trunk, and (c) horizontal rupture of the internal layer of aorta was noticed 6.5cm from the junction of ascending aorta and descending aorta

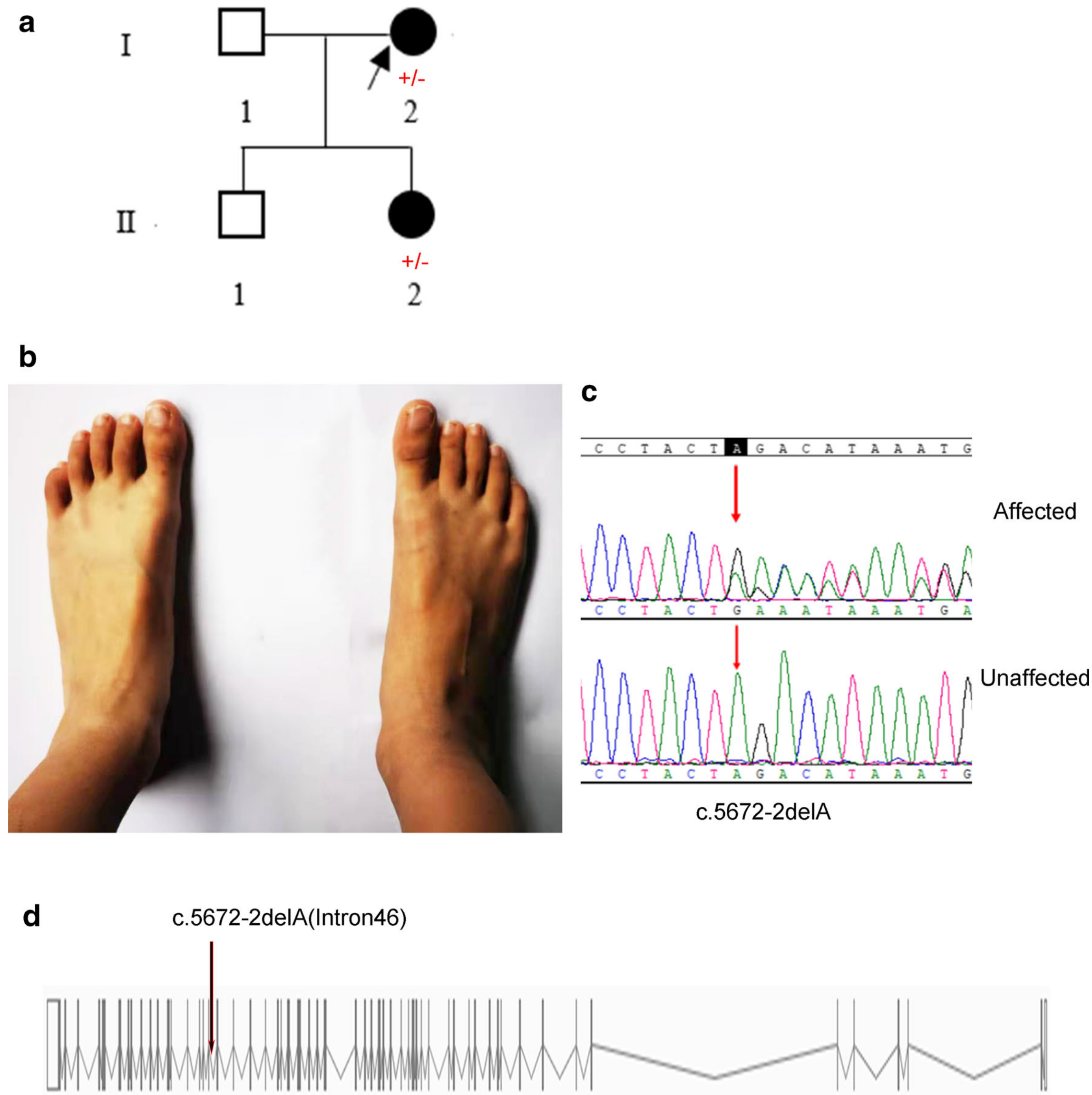


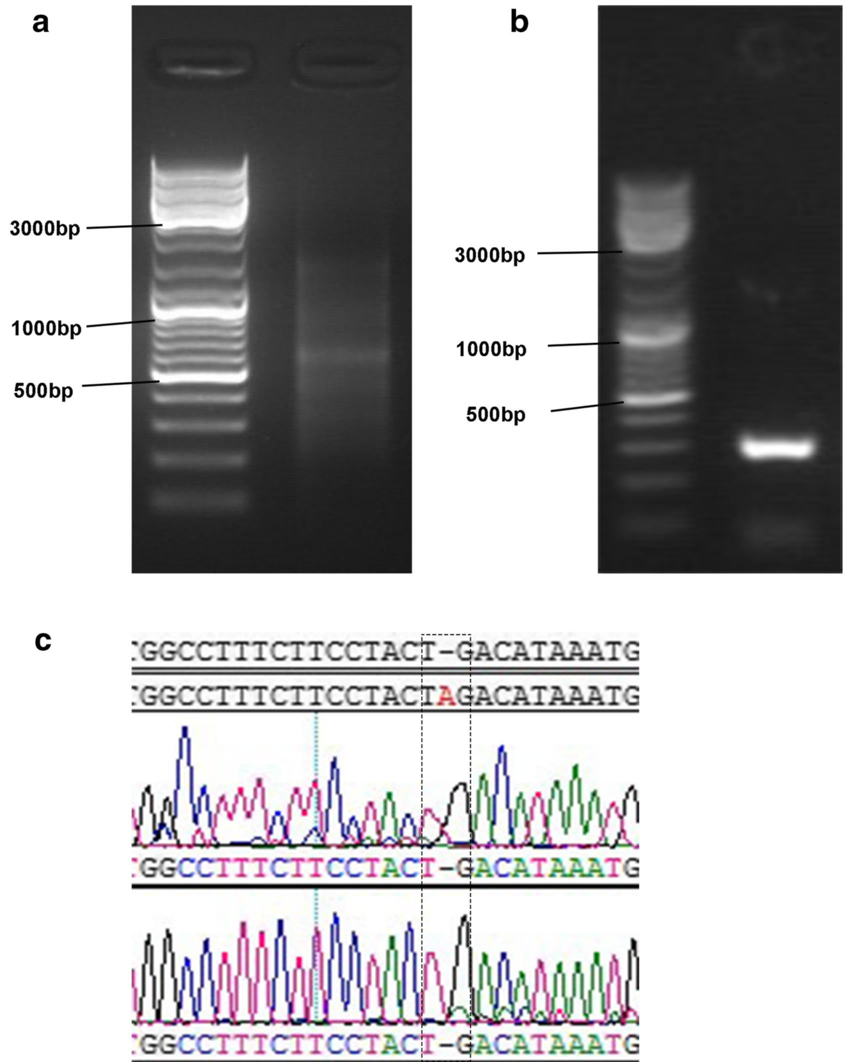
Fig. 2 **a** Pedigree showing two affected members. +/- denotes heterozygous mutation in *FBN1*. **b** Foot features of affected girl showing arachnodactyly. **c** Chromatogram of *FBN1* heterozygous splicing mutation. **d** Genomic organization of *FBN1* and mutation in intron46

Discussion

As we all know, in 1896, MFS was originally reported by Antoine-Bernard Marfan in medical literature. In 1991, heritable mutations of the fibrillin-1 (FBN1) gene were firstly confirmed as a cause of MFS [9]. According to recent studies, genetic factors were critical in the progress of MFS [6]. It is a common view that almost all the MFS can attribute to the mutation in FBN1 gene [8]. All pathogenic mutations within FBN1 can be subdivided into haplo-insufficient (HI) mutations or dominant-negative (DN) mutations [10]. There is a long-standing debate about whether FBN1 mutations lead to MFS through a lack of haploid, dominant-negative mechanisms, or a combination of the two; recent studies have shown that both mechanisms can cause MFS [11]. Usually, HI mutations result in the production of small amounts of normal functional fibrin-1 protein, as mutations can lead to protein degradation and decay, blocking protein translation, mainly through premature translational-termination codon or shifts code mutation to achieve [12]. DN mutations result in structural changes in the firm mutant fibrillin-1 protein or the formation of shorter fibrillin-1 proteins, including missense or exon skip mutations [13].

The variants in FBN1 could constitute abnormal microfibril, resulting in malformation of fibrillin and thus affecting the structure and function of connective tissues. It is reported that about a quarter of the MFS cases are attributed to de novo mutations in FBN1, whereas, when it comes to splicing mutations, the rate was 18.5% [4, 14]. Splicing mutations may result in exon skipping or alter the splice

Fig. 3 **a** RNA electrophoresis of the total RNA of the heart tissue, **b** agarose gel electrophoresis of the PCR products, and **c** the c.5672-2delA was also detected in the PCR products



positions. In most cases, the exon skipping does not result in subsequent sequence changes, only mutant fibrillin-1 with a deletion integration domain, whereas in a few cases, due to nonsense-mediated decay of the mutant transcript, the exon skipping frameshift results in the early appearance of the terminator and the production of less mRNA. The use of a cryptic splice site usually leads to the deletion of amino acids in fibrillin-1 and in a few cases can also result in insertions [15]. For example, a mutation in the donor splice site in intron27 leads to merge entire 111-bp with mRNA, in which case the introns contain a premature stop codon (PTC) that forms a low-expression truncated mutant protein [16]. The intron 46 G+5 to A mutation, which altered the mammalian splice donor consensus, results in exon skip [17].

To our knowledge, the novel mutation c.5672-2delA detected in our case has not been reported, and this mutation is extremely minimal in the normal population. The mutation c.5672-2delA was pathogenic predicted by Mutation Taster online software (http://www.mutationtaster.org). The mutation was detected in the descendent and her daughter with positive clinical characteristics, which support the pathology of this novel mutation. The functional validation of the mutation showed the base deletion at the same site in the PCR products using cDNA as a template. The intron where the mutation is located is retained in the mRNA. It is suggested that this mutation may cause abnormal spliceosome during transcription and may encode abnormal protein.

Sudden aortic death (SAD) is common in forensic practice, and it is well known the relationship between sudden death and TAAD. Genetic mutations are often associated with TAAD [18]. TAAD-related diseases are mainly autosomal dominant, and the likelihood of first-degree relatives is as high as 50%, which may be extremely important for the families of the deceased [19]. And in 2017, Han et al. recommend genetic analysis in SAD cases [20]. Postmortem genetic analysis is important to determine whether a potential genetic disorder is a potential cause of death [21–24]. Although a considerable number of variants associated with MFS have been found through clinical studies, rather small numbers of reports on postmortem genetic analysis among sudden deaths in MFS patients, these mutations supposed to be prone to sudden death in MFS patients, thus providing us with an opportunity to improve the prognosis of carriers [21, 23]. Considering these studies, we performed postmortem genetic analysis and found the underlying gene mutation and carrier; the early diagnosis of MFS in her daughter may lead to medical interval in time and avoid such tragedy.

In conclusion, a novel heterozygous mutation c.5672-2delA, in the FBN1 gene, was likely pathogenic associated with MFS. Through the inspection to the deceased's family, the mutation was found in her daughter with positive physical characteristics. The present work enriches the profile mutations in FBN1 associated with MFS and stresses the importance of postmortem genetic analysis in such cases.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Informed consent Written informed consent for publishing this scientific report was obtained from the direct relative of the decedent in this case.

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